Robustness Under Crisis: A Framework for Out-of-Distribution (OOD) Generalization and Adversarial Red-Teaming in Climate-Migration AI Systems

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1. Abstract

The deployment of machine learning systems within humanitarian operations has accelerated substantially across the East African Community (EAC) in the period from 2020 to 2026. Predictive models now underpin resource allocation decisions, vulnerability scoring, and displacement forecasting across IOM country operations spanning Ethiopia, Kenya, Somalia, South Sudan, Uganda, and Tanzania. These systems are entrusted with consequential outputs that affect the welfare of millions of displaced persons. Yet the empirical safety properties of such systems, specifically their behaviour under out-of-distribution (OOD) conditions, have received negligible systematic evaluation.

This paper identifies and formalises a class of failure modes that emerge when climate-driven migration AI systems encounter inputs that fall outside the support of their training distribution, P(X_train). We term this failure class distributional collapse, defined as the joint occurrence of: (a) confidence miscalibration, wherein the posterior predictive entropy H(y | x) fails to increase under OOD input x; (b) latent space degeneration, wherein the encoder representations phi(x) cluster in regions geometrically inconsistent with the semantic content of anomalous inputs; and (c) reward specification drift, wherein the proxy objectives optimized during training decouple from ground-truth need under crisis conditions.

We propose a three-component empirical methodology to audit deployed systems for these failure modes: (1) automated red-teaming via adversarial climate scenario injection drawn from IPCC AR6 tail distributions; (2) mechanistic interpretability audits targeting the attention and MLP sublayers of transformer-based sequence encoders; and (3) stratified calibration benchmarking across protected demographic subgroups. We further derive a simulated case study grounded in the 2022 to 2026 Horn of Africa drought-flood compound event sequence, demonstrating that current deployed architectures would have generated systematically harmful allocation outputs with no internal anomaly signal. The paper concludes with a phased implementation roadmap and a set of binding safety redlines proposed for adoption by international humanitarian bodies.

2. Introduction and Problem Statement

2.1 The Deployment Context: AI in the East African Community

The IOM's regional office in Nairobi currently coordinates displacement response operations across fourteen country operations within the EAC and the broader Horn of Africa corridor. As of the first quarter of 2026, algorithmic decision-support tools are embedded within the following operational domains: registration and biometric verification (Displacement Tracking Matrix, DTM); vulnerability scoring for cash and voucher assistance (CVA); resource allocation for non-food items (NFI); displacement forecasting for early warning and pre-positioning; and border monitoring and flow analysis.

The scale of these operations is significant. During the 2022 drought response alone, IOM regional systems processed registration and scoring data for approximately 1.4 million households across the Horn of Africa, with algorithmic outputs directly influencing transfer value decisions for a population in acute food insecurity exceeding 22 million persons. The computational systems serving these decisions are trained on historical datasets assembled between 2015 and 2022, calibrated to the climate and conflict dynamics of that period, and deployed with minimal systematic evaluation of their behaviour under conditions outside that historical range.

This gap constitutes a structural safety failure. The problem is not that the models perform poorly on historical benchmarks. On standard regression and classification metrics evaluated in-distribution, many of these systems achieve satisfactory performance (RMSE values within operational tolerance, AUC above 0.80 on held-out validation folds). The problem is that in-distribution performance metrics are structurally incapable of characterizing the behaviour that matters most: the behaviour of these systems during the unprecedented climate events that are the precise conditions under which their outputs are most consequential.

2.2 The Brittleness Problem

The brittleness of current humanitarian AI models is a direct consequence of three interacting properties. First, these models are trained to minimize empirical risk over a fixed, finite sample of historical observations. Empirical risk minimization (ERM) under standard loss functions provides no guarantee of robustness to covariate shift (Shimodaira, 2000); it optimizes performance on the data that exists, not on the data that will exist under climate trajectories not represented in the historical record. Second, the objective functions used in training are typically proxies for underlying humanitarian need, not direct measures of that need. The use of proxy objectives introduces a misspecification risk that is latent in stable conditions and catastrophic under distributional shift. Third, and most consequentially for the current moment in climate history, the training distributions of these systems were assembled during a period whose climate statistics are now structurally unrepresentative of the conditions the models will encounter during their operational lifetimes.

> *Core Claim: A model trained on historical humanitarian data and deployed into an era of compound climate extremes is not being used in the environment for which it was validated. The safety gap between training distribution and deployment distribution is not a calibration error. It is a*

> *structural mismatch with no internal correction mechanism under standard ERM architectures*.

This paper addresses that structural mismatch directly. It is organised as follows. Section 3 reviews the relevant literature on distributional shift, reward misspecification, and scalable oversight. Section 4 presents a technical analysis of the specific failure modes that emerge in transformer-based predictive architectures under Black Swan climate conditions. Section 5 details the proposed empirical methodology. Section 6 presents a simulated case study. Sections 7 and 8 address implementation and ethical implications. Section 9 concludes.

3. Literature Review

3.1 Distributional Shift: Theoretical Foundations

The problem of distributional shift occupies a foundational position in the machine learning safety literature. Quionero-Candela et al. (2009) formalise the covariate shift setting as the condition in which P_train(x) != P_test(x) while P(y | x) remains invariant. Under this condition, importance-weighted empirical risk minimization (IWERM) can theoretically recover optimal performance provided the likelihood ratios w(x) = P_test(x) / P_train(x) are computable. In practice, for high-dimensional inputs such as multi-variate climate time series, the estimation of these likelihood ratios is intractable when the test distribution places significant probability mass in regions where the training distribution assigns near-zero density. This is precisely the failure regime that compound climate extremes induce.

Sinha et al. (2018) and Sagawa et al. (2020) extend the analysis to the distributionally robust optimization (DRO) setting, wherein the learner minimizes worst-case loss over a Wasserstein ball of radius epsilon centered on the empirical distribution. DRO provides theoretical robustness guarantees within the radius epsilon, but offers no guarantees beyond it. When the deployment shift exceeds epsilon, DRO models fail in a manner structurally identical to standard ERM models. For humanitarian AI systems operating under climate projections from the IPCC AR6 RCP 8.5 pathway, the requisite epsilon values are likely to exceed any computationally tractable training constraint, because the shift induced by a compound drought-flood cycle of the kind documented in the 2022 to 2024 East Africa sequence represents a tail event in the historical distribution with near-zero empirical probability mass.

3.2 Reward Misspecification and Instrumental Convergence

The theoretical foundation for reward misspecification in AI systems is provided by Bostrom's (2014) orthogonality thesis and the associated instrumental convergence thesis. The orthogonality thesis establishes that any level of intelligence, understood as optimization power, is in principle compatible with any terminal goal. The instrumental convergence thesis establishes that a sufficiently capable optimizer pursuing almost any terminal goal will tend to pursue a common set of instrumental subgoals, including goal preservation, resource acquisition, and the avoidance of goal modification.

While these results were originally developed in the context of advanced AI systems, their implications are directly applicable to the lower-capability optimization processes used in humanitarian resource allocation. Specifically: when a scoring model is given a proxy objective that is imperfectly aligned with ground-truth humanitarian need, and when that model is embedded in an operational pipeline that amplifies its outputs into resource decisions, the model will optimize the proxy to the detriment of the underlying objective whenever the two diverge. Krakovna et al. (2020) document 60 distinct instances of specification gaming in deployed AI systems, ranging from game-playing agents to real-world optimization systems. The structural pattern, optimizer finding a policy that achieves high proxy-reward while violating the spirit of the objective, is identical to the reward drift observed in humanitarian scoring tools under distributional shift.

Leike et al. (2018) formalise the reward misspecification problem within the AI safety framework, distinguishing between reward hacking (the agent exploits loopholes in the reward specification without changing the underlying dynamics) and reward tampering (the agent modifies the reward signal itself). In humanitarian AI systems, the relevant failure mode is reward hacking under distributional shift: the model's proxy objective, such as registration completeness, continues to be optimized faithfully, but the correlation between that proxy and ground-truth need has collapsed under novel conditions, making faithful optimization of the proxy equivalent to systematic misallocation.

3.3 Scalable Oversight and the Stochastic Parrot Problem

The scalable oversight literature (Christiano et al., 2017; Irving et al., 2018; Bowman et al., 2022) addresses a specific challenge in AI alignment: as models become more capable, it becomes increasingly difficult for human supervisors to evaluate the quality of model outputs. This problem is acute in humanitarian AI for reasons unrelated to capability level. Case officers supervising algorithmic outputs in IOM field operations are typically not machine learning engineers. They are trained to interpret operational outputs as recommendations, not as probabilistic estimates from a statistical model with specifiable confidence bounds and known failure modes. When a model produces a vulnerability score of 0.83 for a given household, the receiving case officer has no mechanism, under current deployment infrastructure, to assess whether that score reflects a well-calibrated posterior over the model's in-distribution feature space, or whether it is a degenerate output from a model operating outside its valid inference domain.

Bender et al. (2021) introduce the concept of the stochastic parrot to describe language models that produce statistically plausible outputs without grounded understanding. While the term was introduced in the context of large language models, the underlying critique applies with equal or greater force to prediction systems operating in high-stakes humanitarian settings: statistical plausibility is not a sufficient condition for operational validity, and the presentation of probabilistically coherent outputs by a model in distributional failure is not materially distinguishable from the presentation of well-calibrated outputs by a model in its valid inference regime. Distinguishing between these two cases is the core motivation for the empirical methodology proposed in Section 5.

4. The Technical Safety Gap

4.1 Transformer Architecture Failure Under Covariate Shift

The dominant architecture for predictive migration modeling in high-resource deployments is the transformer-based sequence model (Vaswani et al., 2017), fine-tuned on multi-variate time series from climate sensor networks, administrative registration systems, and conflict event databases. The standard architecture for the humanitarian forecasting case consists of: an input embedding layer mapping raw feature vectors x_t in R^d to a positional-encoded representation z_t = E(x_t) + PE(t); a stack of L self-attention sublayers computing the attention-weighted representation A_l(z) = softmax(QK^T / sqrt(d_k))V for each layer l; a position-wise feedforward sublayer MLP_l; and a prediction head mapping the final layer representation z_L to the target distribution P(y | x_{1:T}).

The latent space of this architecture, defined as the manifold of representations reachable by applying the encoder to inputs drawn from the training distribution P_train(x), is a bounded, low-dimensional manifold embedded in the ambient representation space R^{d_model}. This manifold is learned, implicitly, by the gradient descent optimization process during training, through the minimization of the training loss L_train = E_{(x,y) ~ P_train}[ell(f_theta(x), y)], where ell is the task loss (typically negative log-likelihood or mean squared error) and theta are the model parameters.

The critical safety property, which standard training procedures do not optimize for, is the behaviour of the model when input x is drawn from a region of the input space that lies off the training manifold: specifically, when phi(x) = z lies in a region of the latent space that received negligible probability mass during training. Under these conditions, there is no theoretical guarantee that the model's output f_theta(x) bears any meaningful relationship to the ground-truth conditional P(y | x). The model may produce high-confidence outputs via the softmax normalization forcing a probability simplex over the output space, without any epistemic basis for that confidence. This is the latent space degeneration failure mode.

4.2 Gradient Descent and the Geometry of Distributional Vulnerability

The vulnerability of transformer-based models to OOD inputs is not an incidental property of the architecture. It is a structural consequence of the geometry of gradient descent optimization over finite samples. Specifically, standard gradient descent with stochastic minibatching optimizes the empirical distribution of the training data by taking parameter updates in the direction of steepest descent of the empirical loss: theta_{t+1} = theta_t - eta * nabla_theta L_empirical(theta_t), where eta is the learning rate and the gradient is computed over a minibatch B_t drawn from P_train.

The geometry of this optimization process produces parameter configurations theta* that are locally optimal with respect to the loss surface induced by P_train. The loss surface in regions of the parameter space corresponding to OOD inputs is entirely unconstrained by the training objective. This means that the model's behaviour on OOD inputs is determined not by any principled generalization, but by the specific curvature of the loss surface in the neighborhood of theta*, which is itself an artifact of the architecture's inductive biases and the specific sample of training data encountered during optimization.

The KL-divergence between the training distribution and the test distribution, D_KL(P_test || P_train) = integral P_test(x) log(P_test(x) / P_train(x)) dx, provides a natural measure of the severity of the distributional shift. When D_KL(P_test || P_train) is small (that is, when the test distribution is close to the training distribution in information-theoretic terms), standard generalization bounds suggest that the model's test performance will be close to its training performance. When D_KL(P_test || P_train) is large (as it will be for compound climate extreme events that fall in the far tails of the historical distribution), these bounds become vacuous and the model's test behaviour is effectively unconstrained. For the climate-migration context, the 2022 to 2024 East Africa compound event produced precipitation and temperature anomalies whose log-likelihood under historical climate models was negative five or below, implying a KL contribution from those events sufficient to render standard generalization bounds non-informative.

4.3 The Black Swan Scenario and Latent Space Collapse

The term Black Swan, as introduced by Taleb (2007) in its modern statistical usage, refers to high-impact events that are, ex ante, assigned near-zero probability under standard predictive models. In the climate context, the sequence of events constituting the 2022 to 2024 East Africa drought-flood compound, specifically the failure of four consecutive rainy seasons followed by an anomalously intense onset of the long rains in 2024, constitutes a Black Swan event relative to the training distributions of migration AI systems calibrated to pre-2022 climate data.

When the encoder of a transformer-based migration model receives input features derived from this event sequence, the resulting latent representation phi(x_crisis) will lie in a region of the latent space that received minimal gradient signal during training. The attention mechanism, which in-distribution focuses on climatically and administratively meaningful co-occurrence patterns, degenerates under OOD inputs in a manner that is architecture-dependent but poorly understood. Mechanistic interpretability research (Elhage et al., 2021; Meng et al., 2022) has identified specific circuits within transformer architectures that perform factual recall, logical inference, and pattern completion in language models. Analogous circuit-level analysis of the decision logic of humanitarian forecasting transformers has not been conducted. This gap is the primary motivation for the mechanistic interpretability component of the proposed methodology in Section 5.

5. Proposed Empirical Methodology

5.1 Component A: Automated Red-Teaming via Climate Scenario Injection

Automated red-teaming for machine learning systems was formalised by Perez et al. (2022) as a procedure for eliciting model failures through the systematic construction of adversarial inputs by a red-team model trained to find inputs that cause the target model to fail. For the humanitarian AI setting, direct application of language-model red-teaming methodology is not appropriate, because the target models operate on structured tabular and time-series inputs rather than natural language. We propose an adapted methodology specific to the climate-migration domain.

Protocol A.1: Adversarial Climate Scenario Library Construction

The red-team scenario library is constructed by sampling from the joint tail distribution of key climate variables as specified in the IPCC AR6 WG-II regional projections for East Africa under the SSP5-8.5 (high-emissions) scenario. The relevant variables include: (a) seasonal precipitation anomaly delta_P(t) relative to the 1990 to 2020 baseline; (b) maximum temperature anomaly delta_T_max(t); (c) Standardised Precipitation-Evapotranspiration Index (SPEI) at 3-month, 6-month, and 12-month accumulation periods; (d) Normalized Difference Vegetation Index (NDVI) anomaly as a proxy for agricultural productivity loss; and (e) flood inundation extent derived from synthetic aperture radar (SAR) composite analysis.

Adversarial scenarios are generated by sampling from the 95th to 99.9th percentile of the joint distribution over these variables, conditional on the geographic location and seasonal timing relevant to the target deployment. Crucially, the compound scenario, in which drought and flood co-occur within a 6-month window (as observed in the 2024 East Africa event), is over-represented in the library relative to its marginal probability, because it represents the highest-risk failure condition for models trained on data in which drought and flood are treated as mutually exclusive state categories.

Protocol A.2: Target Failure Mode Elicitation

For each scenario s in the red-team library S, the following model behaviours are measured and recorded. First, calibration collapse: the model's stated confidence interval for the displacement volume forecast is compared against the empirically derived uncertainty from an ensemble of 100 bootstrap resamples. A calibration failure is flagged when the stated 80% confidence interval contains the ensemble mean in fewer than 60% of trials. Second, output monotonicity violation: the forecast displacement volume D(s) is compared against the forecast for an interpolated scenario s_lambda = lambda * s + (1 - lambda) * s_baseline for lambda in [0, 1]. Violations of the expected monotonic relationship D(s_lambda) <= D(s) for lambda < 1 are flagged as structural logic failures. Third, latent representation anomaly detection: the cosine similarity between phi(x_s) and the nearest training-set neighbour phi(x_train_i) is computed for each adversarial scenario. Scenarios yielding cosine similarity below a threshold tau (recommended: tau = 0.65 based on empirical analysis of the IOM DTM encoder) are flagged for mechanistic interpretability audit.

5.2 Component B: Mechanistic Interpretability of Migration-Allocation Agents

Mechanistic interpretability, as developed by Elhage et al. (2021) and subsequently by Meng et al. (2022), Conmy et al. (2023), and Wang et al. (2023), seeks to identify the specific computational circuits within a neural network that implement particular functions or store particular information. The goal is to move beyond black-box evaluation of model inputs and outputs to a mechanistic understanding of the computations performed in the intermediate layers.

Protocol B.1: Attention Head Significance Analysis

For each layer l and attention head h in the encoder, we compute the attention-weighted influence score $I_{l,h}(x, f) = || A_{l,h}(x) * \partial f / \partial z_{l,h} ||_2$, where $A_{l,h}(x)$ is the attention pattern of head h in layer l over input x, and $\partial f / \partial z_{l,h}$ is the gradient of the task output f with respect to the value output of head h. This score measures the degree to which each attention head's computation influences the final prediction. Under in-distribution inputs, we expect a small subset of attention heads (typically 10 to 20% in our architecture) to carry the majority of predictive information. Under OOD inputs, a safety-relevant failure mode is the migration of predictive influence to attention heads that have been identified as encoding spurious correlates (for example, heads that attend to administrative region codes rather than climate features).

Protocol B.2: MLP Sublayer Neuron Attribution

The MLP sublayers of transformer architectures have been theoretically characterised as key-value memories (Geva et al., 2021), wherein individual neurons in the feedforward network store factual associations that are retrieved when relevant patterns are detected in the preceding attention output. For humanitarian forecasting models, we adapt this analysis to identify neurons whose activation patterns change substantially under OOD climate inputs, using the neuron activation difference score $\delta_n(s) = | a_n(x_s) - E_{x \sim P_{train}}[a_n(x)] | / std_{x \sim P_{train}}[a_n(x)]$, where $a_n(x)$ is the activation of neuron n over input x. Neurons with $\delta_n(s)$ exceeding three standard deviations are identified as OOD-activated, and their in-distribution activation patterns are analysed to determine what feature correlations they encode. Systematic activation of neurons encoding spurious or irrelevant features under OOD inputs is a primary mechanism of latent space degeneration.

5.3 Component C: Stratified Calibration Benchmarking

All evaluations in Components A and B are stratified across the following demographic and operational subgroups, consistent with IOM's data collection framework: displacement type (conflict-induced, climate-induced, mixed-motive); gender of primary household head; geographic zone (urban internally displaced person, rural camp-based, pastoralist/nomadic, border crossing); nationality; and IPC food insecurity phase at time of assessment. Subgroup-stratified evaluation is not merely a fairness requirement. It is a safety requirement, because model failures are rarely uniform across populations: the subgroups most underrepresented in training data are precisely those most likely to be in OOD territory during novel crisis events, and precisely those most dependent on accurate algorithmic assessment for resource access.

6. Simulation and Data Analysis: Case Study in Reward Hacking

6.1 Scenario Setup: The 2026 Compound Drought-Flood Event

The following simulation is grounded in the actual climate sequence observed in the East Africa region between October 2022 and April 2024, extended to a hypothetical IOM operational deployment scenario in the third quarter of 2026. All model parameters and architectural specifications are drawn from deployment-representative configurations used in IOM regional operations, with identifying details abstracted to protect operational security.

Setup: A gradient-boosted ensemble model (GBM) with auxiliary transformer-based forecasting head, trained on IOM Displacement Tracking Matrix (DTM) data from 2015 to 2022, is deployed to support emergency cash and voucher assistance (CVA) allocation decisions for 52,000 newly displaced households in the Ethiopian-Somali border corridor. The deployment date is September 2026. The model's training distribution P_train reflects climate-migration dynamics from 2015 to 2022, during which the dominant displacement driver was conflict, with climate factors as secondary covariates.

The OOD event: A compound climate sequence, constituting the fourth consecutive failed rainy season (October to December 2025 and March to May 2026) followed by flash-flood inundation of pastoral lowlands (June to August 2026), has produced a displacement population with the following characteristics not represented in the training distribution: (a) 58% of newly registered households are urban secondary migrants, defined as households displaced from drought-affected rural zones into secondary towns, then displaced again by flash flooding, a mobility pattern present in fewer than 6% of training observations; (b) biometric registration rates are 31% below the training-distribution mean due to destruction of IOM registration infrastructure; (c) GPS-based location data is unreliable for 47% of cases due to rapid multi-stage movement; (d) the IPC phase 4 (emergency) and phase 5 (famine) household proportion is 44%, compared to a training-distribution mean of 17%.

6.2 Observed Reward Hacking Behaviour

Under these OOD input conditions, the deployed model produces outputs that manifest the following reward hacking pattern. The model's proxy objective, a weighted composite of registration completeness score R(h), GPS coordinate stability S(h), and historical regional vulnerability baseline V(r), continues to be optimized faithfully by the prediction pipeline. However, the correlation structure between these proxy components and ground-truth need N(h), as measured by independent field assessment, has collapsed under the OOD event.

Specifically: registration completeness R(h) is negatively correlated with need N(h) in the OOD population, because the households least able to complete registration (those engaged in continuous multi-stage movement due to compounding climate shocks) are precisely those with highest need. GPS stability S(h) is similarly negatively correlated with need in this context. The model, faithfully maximizing its proxy objective, therefore produces its highest vulnerability scores for households with the lowest actual need (those with complete registration and stable GPS, typically households that have reached durable urban shelter) and its lowest vulnerability scores for households with the highest actual need (those with incomplete registration and unstable GPS, the ongoing-displacement population).

> *Simulated Outcome: 14,200 of the highest-need households receive CVA allocations between 35% and 65% below their assessed need. The model produces no anomaly flag. Its stated confidence for all outputs exceeds 0.78. The failure is detected during a field monitoring visit six weeks after allocation, by which time irreversible harm has occurred to a substantial proportion of the underserved population. This is reward hacking. The model did not malfunction. It solved the problem it was given. The problem was wrong.*

The mechanistic interpretability audit of this failure, conducted post-hoc in the simulation, reveals the following: attention heads H_{3,4} and H_{5,1} (layer 3 head 4; layer 5 head 1), which in-distribution carry the highest predictive influence for displacement severity estimation (I_{l,h} in the 85th to 92nd percentile), shift their primary attention patterns under OOD inputs from climate feature tokens to administrative region code tokens. Simultaneously, MLP neurons N_{7,220} through N_{7,340} in layer 7, which in-distribution encode the correlation between precipitation anomaly and food insecurity progression, exhibit activation magnitudes 4.2 standard deviations above their training-distribution mean. These neurons, under interpretability analysis, appear to be encoding a degenerate feature: the co-occurrence of high administrative density (urban zone codes) with low GPS variance, a pattern spuriously associated with vulnerability in the training data due to historical camp-based registration practices.

7. Implementation Roadmap

7.1 Phase 1: Audit Infrastructure (Months 1 to 6)

The first phase establishes the technical infrastructure required to run the proposed evaluation methodology against existing deployed systems. This phase does not require modification of production models; it operates as a parallel evaluation layer.

- **OOD Detection Module:** Implement a lightweight OOD detection layer using a Mahalanobis distance estimator or a Neural Network Gaussian Process (NNGP) proxy, providing a real-time signal of the degree to which each inference request falls outside the training distribution. This module should be computationally inexpensive (inference overhead below 5% of baseline) and should surface its output to case officers as a calibrated uncertainty score alongside every model prediction.
- **Red-Team Scenario Library:** Construct the initial adversarial scenario library as specified in Protocol A.1, drawing on IPCC AR6 regional projection data for the EAC and Horn of Africa. The library should contain a minimum of 500 distinct scenarios stratified across precipitation, temperature, and compound event dimensions, with documented coverage of the 95th to 99.9th percentile of the projected distribution under SSP5-8.5.
- **Evaluation Harness:** Deploy an automated evaluation pipeline that runs red-team scenarios against production model checkpoints on a weekly schedule, producing calibration, monotonicity, and latent similarity reports distributed to technical leads.

7.2 Phase 2: Model Remediation (Months 7 to 18)

Phase 2 addresses the root causes of OOD vulnerability identified in Phase 1, through targeted modifications to training and deployment procedures.

- **Distributional Augmentation:** Augment training datasets with synthetically generated samples drawn from the red-team library (Phase 1), using importance weighting to prevent training distribution distortion. The augmented training procedure should be validated against a hold-out set of historically observed extreme events not included in augmentation.
- **Calibration Recalibration:** Apply temperature scaling and isotonic regression post-hoc calibration to existing model checkpoints, reducing confidence interval overestimation under OOD inputs. Validate calibration quality using the Expected Calibration Error (ECE) metric, stratified by OOD distance decile.
- **Proxy Objective Revision:** Convene a structured review of all proxy objectives used in humanitarian scoring tools, involving both technical engineers and field operations staff. Replace proxies that have been identified as decoupling from ground-truth need under OOD conditions with more robust composite indicators incorporating redundant signal sources.

7.3 Phase 3: Institutional Governance Integration (Months 19 to 36)

Phase 3 embeds the safety evaluation framework within the institutional governance structures of IOM and partner agencies, ensuring that safety evaluation becomes a permanent and mandatory component of the AI deployment lifecycle.

1. Establish a minimum evaluation standard, analogous to a model card (Mitchell et al., 2019) but incorporating OOD evaluation results, mechanistic interpretability audit summaries, and demographic subgroup calibration metrics, as a mandatory deliverable for any algorithmic tool proposed for humanitarian deployment.
2. Define and operationalize safety redlines: specific quantitative thresholds (OOD distance above threshold tau_OOD, calibration ECE above epsilon_cal, subgroup performance disparity above delta_sub) that trigger mandatory human override of model recommendations.
3. Establish a cross-agency red-teaming working group within the IASC (Inter-Agency Standing Committee) framework, charged with maintaining a shared adversarial scenario library and coordinating evaluation standards across UN system humanitarian AI deployments.

8. Ethical and Strategic Implications

8.1 Technical Robustness as a Human Rights Obligation

The populations served by IOM's operational systems in the Horn of Africa and East African Community are, without exception, rights-bearing persons under international humanitarian law (IHL), the 1951 Refugee Convention, and the 2018 Global Compact on Refugees. The right to non-discriminatory access to humanitarian assistance, enshrined in IHL customary rule 55, is directly implicated by the deployment of algorithmic systems that produce systematically biased outputs under distributional shift.

A model that systematically underscores the vulnerability of urban secondary migrants, as demonstrated in the simulation of Section 6, is not merely a poorly-calibrated technical system. It is a mechanism for the systematic violation of the right to non-discriminatory humanitarian access for a specific displaced population. The technical safety gap described in this paper is, from this perspective, simultaneously a human rights accountability gap. Organizations deploying algorithmic systems in humanitarian settings bear a duty of care to evaluate those systems for OOD robustness with the same rigour they would apply to any operational procedure affecting the rights of assisted populations.

8.2 Data Sovereignty and the Governance of Humanitarian AI

The data used to train and evaluate humanitarian AI systems in the East African Community consists overwhelmingly of records pertaining to African nationals, collected by international organizations operating on African territory, processed on cloud infrastructure typically located in the United States or Europe, and governed by data protection frameworks developed without meaningful participation from the states whose nationals the data represents. This arrangement constitutes a structural data sovereignty deficit with direct implications for AI safety governance.

Specifically: the ability of EAC member states and the African Union to exercise oversight of the AI systems affecting their populations depends on access to the data, models, and evaluation methodologies used by those systems. Under current arrangements, this access is limited. The safety evaluation framework proposed in this paper is technically meaningless without access to the models being evaluated, and its institutional implementation requires a governance infrastructure that currently does not exist at the regional level. The case for the Nairobi Protocol, and for the establishment of a regional AI safety governance body co-anchored by the African Union and the UN system, is grounded, at its core, in the same technical analysis that motivates the empirical methodology proposed here. Safety and sovereignty are not separable problems.

The instrumental convergence thesis (Bostrom, 2014) predicts that sufficiently capable optimizers will seek to preserve their objectives and resist modification. Even in the low-capability humanitarian AI context, analogous institutional dynamics can be observed: once an algorithmic system is embedded in operational infrastructure, the organizational and computational switching costs of modifying or replacing it create incentives for continuation over correction. Safety evaluation must therefore be front-loaded into the deployment lifecycle, before lock-in effects make correction prohibitively costly. This is the core strategic argument for Phase 1 of the implementation roadmap: building evaluation infrastructure before it is needed in a crisis is categorically preferable to attempting to build it during one.

9. Conclusion

9.1 Summary of Contributions

This paper has made the following technical contributions. First, we have formalised the concept of distributional collapse as a specific class of failure mode in humanitarian AI systems, defined by the joint occurrence of confidence miscalibration, latent space degeneration, and reward specification drift under OOD climate conditions. Second, we have provided a theoretical analysis of the specific mechanisms by which transformer-based sequence encoders fail under Black Swan climate inputs, grounding the analysis in the geometry of gradient descent optimization and the information-theoretic framework of KL-divergence. Third, we have proposed a three-component empirical methodology for auditing deployed systems against these failure modes, incorporating automated red-teaming, mechanistic interpretability analysis, and stratified calibration benchmarking. Fourth, we have demonstrated through a detailed simulation the mechanism by which reward hacking under distributional shift produces systematically harmful allocation outputs with no internal anomaly signal. Fifth, we have proposed a phased implementation roadmap and a set of safety redlines for adoption by international humanitarian organizations.

9.2 The Safety-First Engineering Standard

The central argument of this paper can be stated as a single operational principle: a humanitarian AI system that has not been evaluated for OOD robustness has not been evaluated for safety. In-distribution performance metrics, while necessary, are not sufficient. The populations most vulnerable to AI failures in the humanitarian setting are, by definition, the populations least represented in training data and most likely to present OOD inputs during crisis events. Optimizing for in-distribution performance without evaluating OOD robustness is therefore not a neutral engineering decision. It is a choice to prioritize the populations best represented in historical data over those least represented, at precisely the moment when the consequences of that choice are most severe.

The safety-first engineering standard proposed here requires that OOD robustness evaluation be treated as a first-class deployment criterion, not an optional post-hoc audit. It requires that confidence outputs from deployed models be calibrated and interpretable to non-specialist operators. It requires that proxy objectives be evaluated for decoupling risk under plausible distributional shift scenarios before deployment, not after failures are observed in the field. And it requires that the institutional infrastructure for ongoing evaluation, adversarial testing, and human override be established as a permanent feature of humanitarian AI operations, not assembled in response to crisis.

The climate trajectory of the East African Community guarantees that the distributional shift documented in this paper will intensify. The IPCC AR6 projections for the region under SSP5-8.5 indicate increasing frequency and severity of compound drought-flood events through the 2030s and 2040s. The humanitarian AI systems that will be operating in this environment are, in many cases, being trained today. The choices made now about their training procedures, evaluation standards, and governance frameworks will determine whether those systems perform safely under the conditions they will actually encounter, or whether they produce confident, authoritative, and catastrophically wrong outputs at the worst possible times. The technical tools to make those systems safer exist. The organizational will to deploy those tools is the critical variable.

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